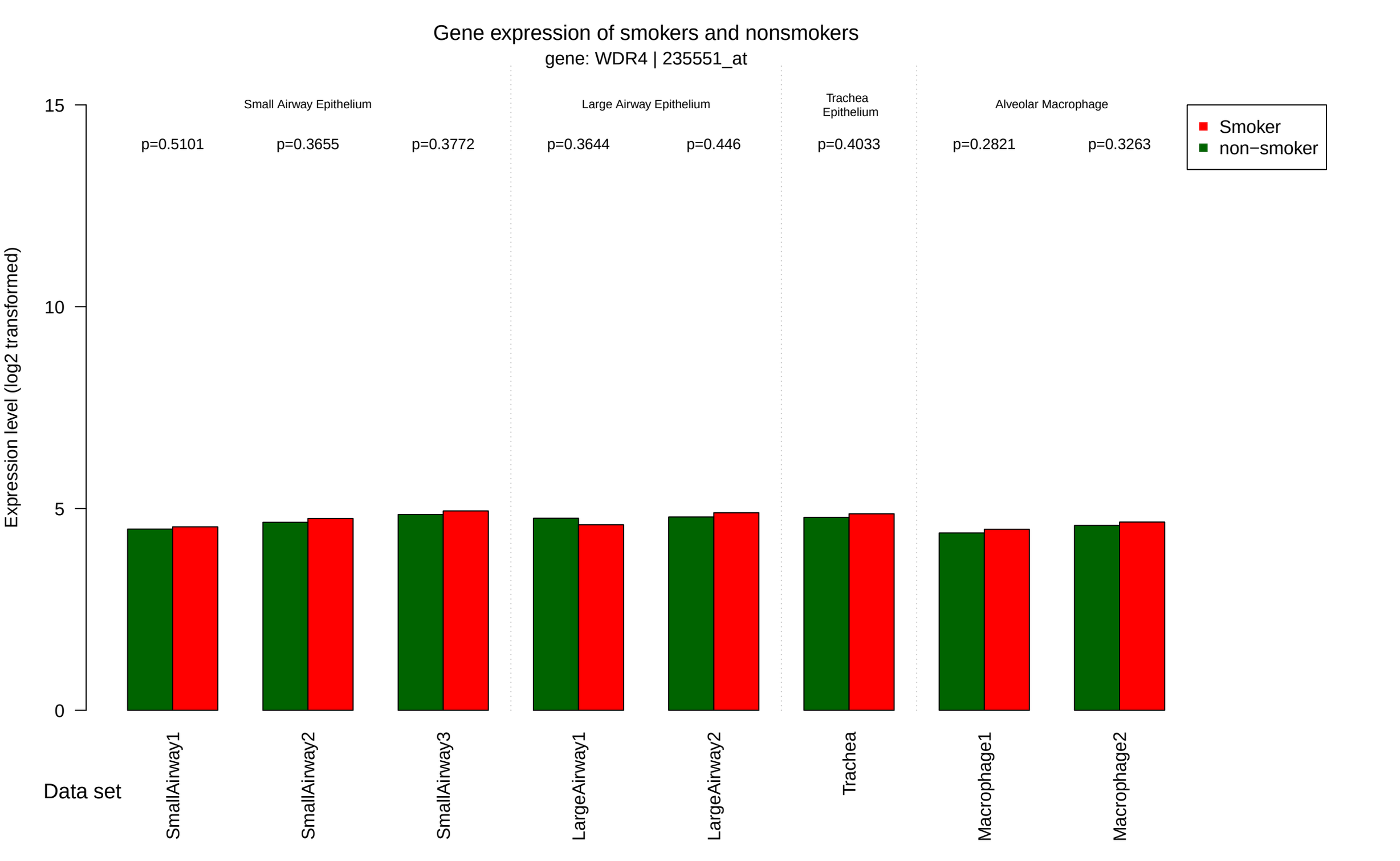
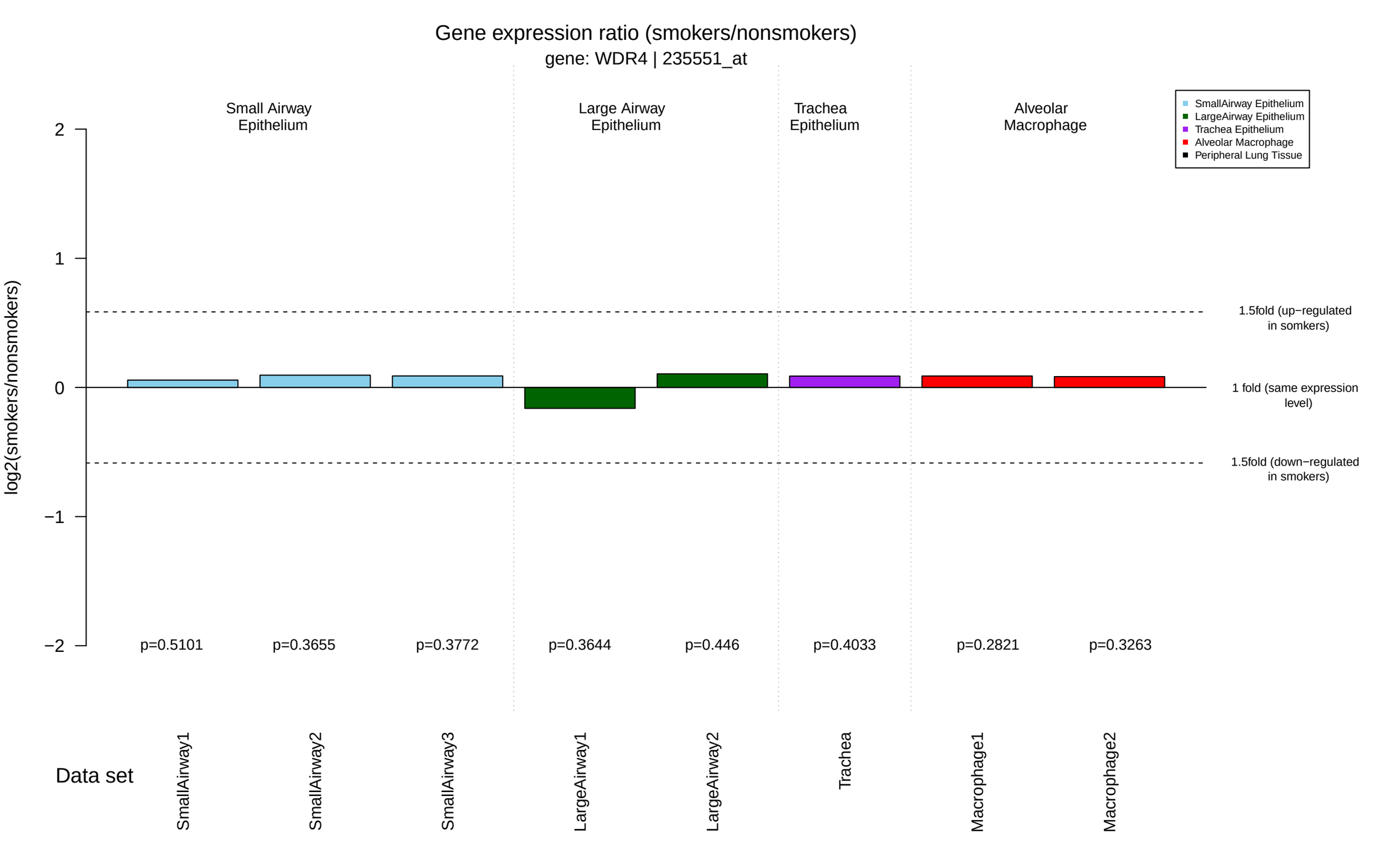
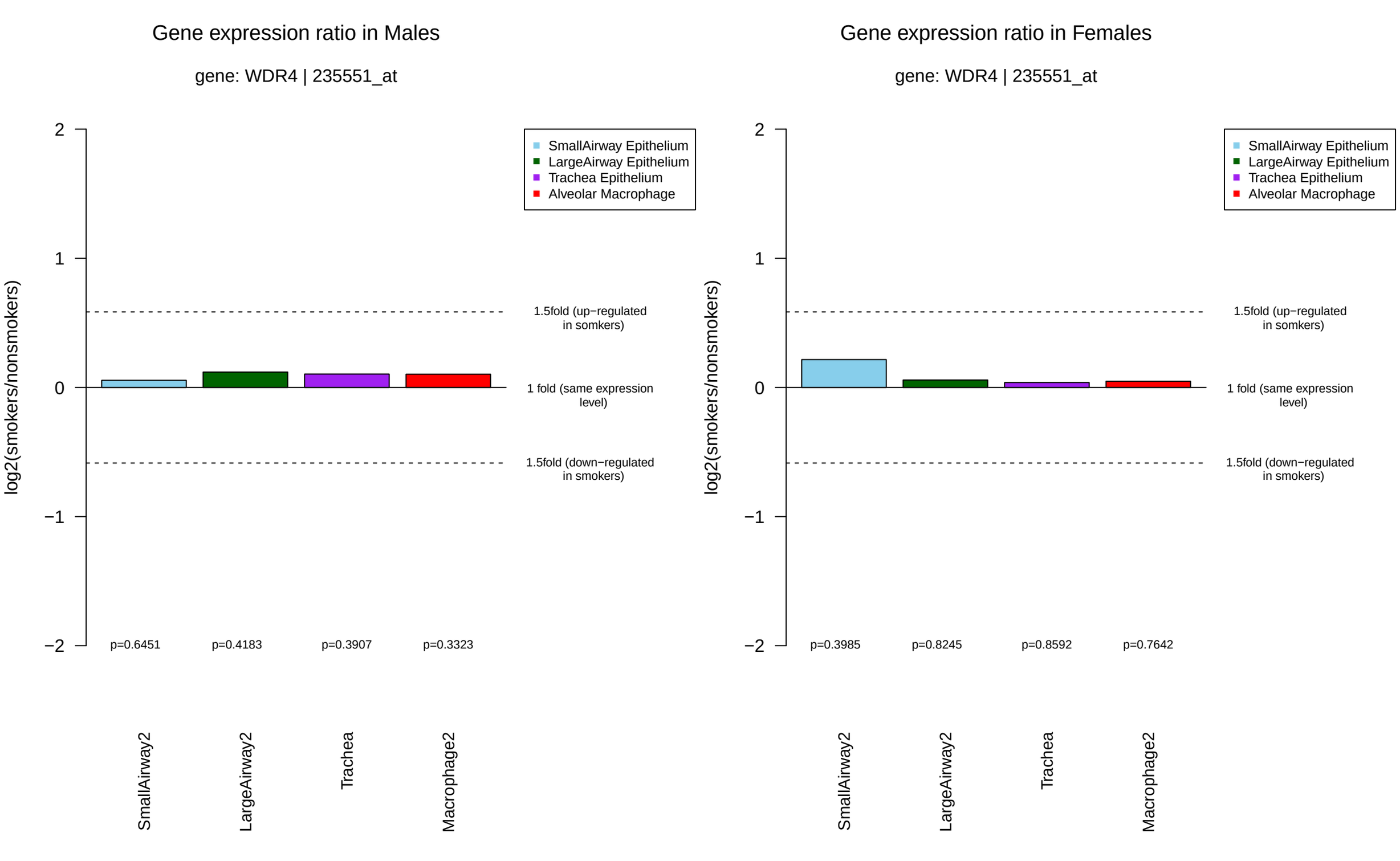
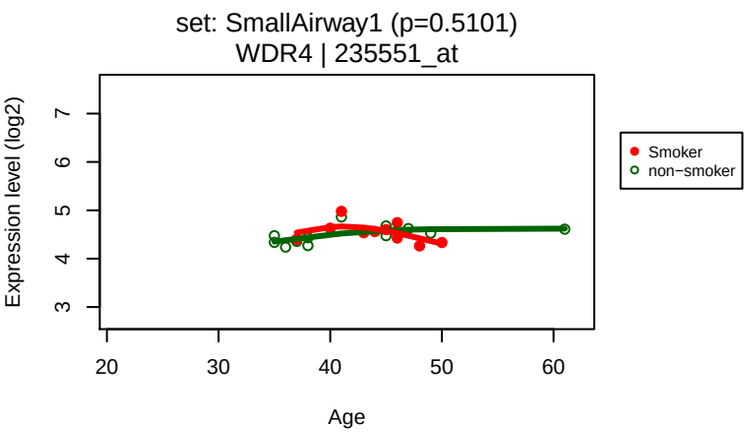
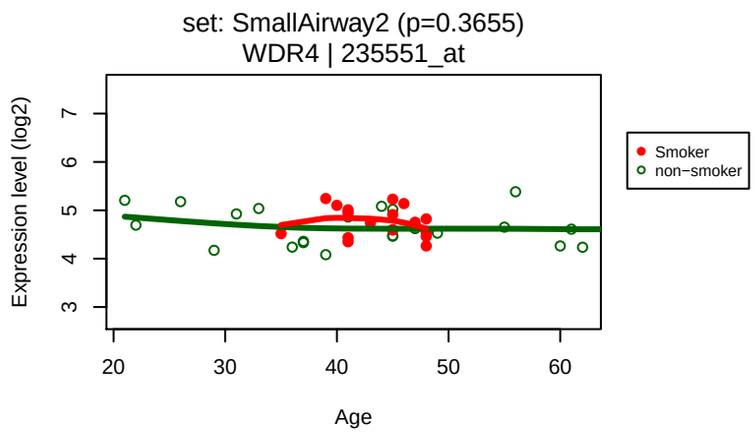
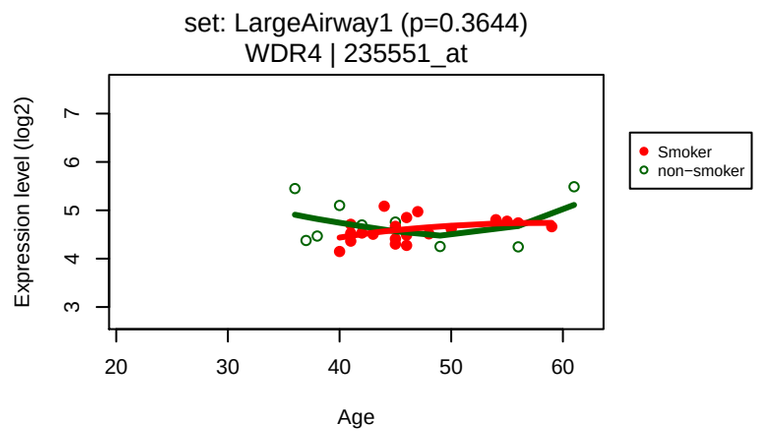

Gene expression of smokers and nonsmokers
gene: WDR4 | 235551_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5101
p=0.3655
p=0.3772
p=0.3644
p=0.446
p=0.4033
p=0.2821
p=0.3263
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

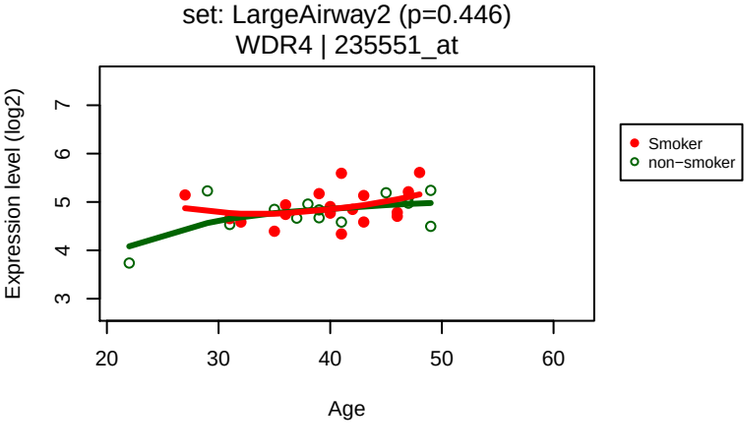

Gene expression ratio (smokers/nonsmokers)
gene: WDR4 | 235551_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5101
p=0.3655
p=0.3772
p=0.3644
p=0.446
p=0.4033
p=0.2821
p=0.3263
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

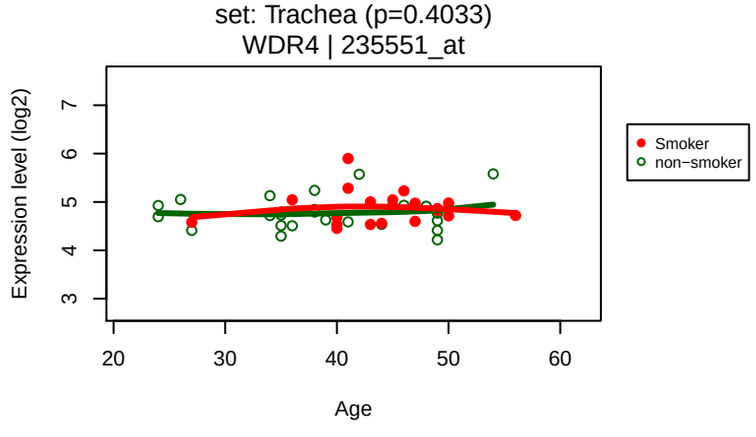

Gene expression ratio in Males
gene: WDR4 | 235551_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6451
p=0.4183
p=0.3907
p=0.3323
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WDR4 | 235551_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3985
p=0.8245
p=0.8592
p=0.7642
SmallAirway2
LargeAirway2
Trachea
Macrophage2

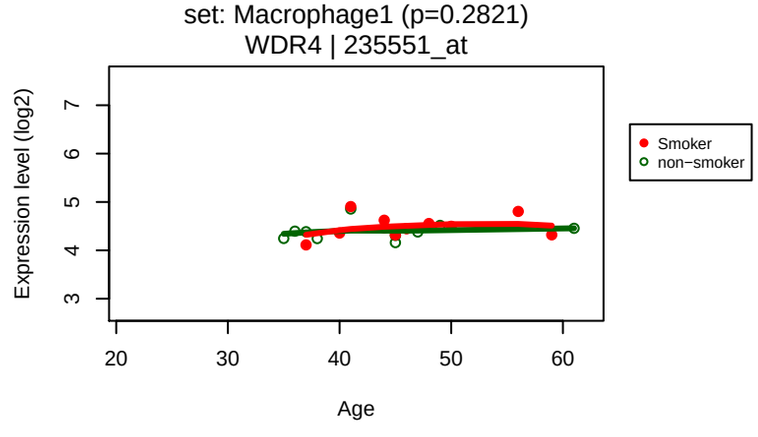

set: SmallAirway1 (p=0.5101)
WDR4 | 235551_at
Expression level (log2)
Age
Smoker
non−smoker

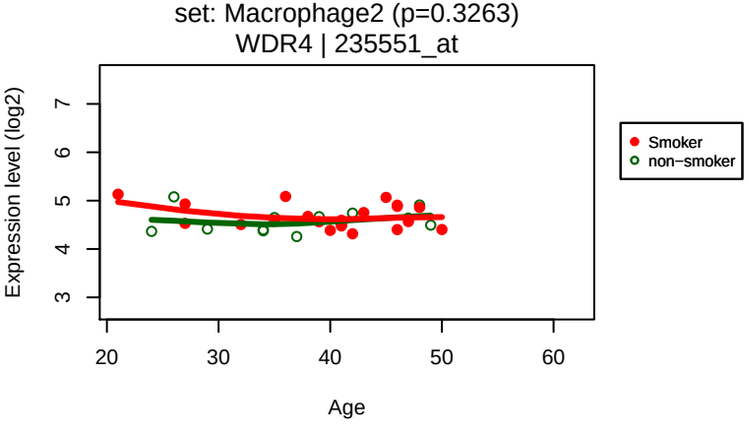

set: SmallAirway2 (p=0.3655)
WDR4 | 235551_at
Expression level (log2)
Age
Smoker
non−smoker

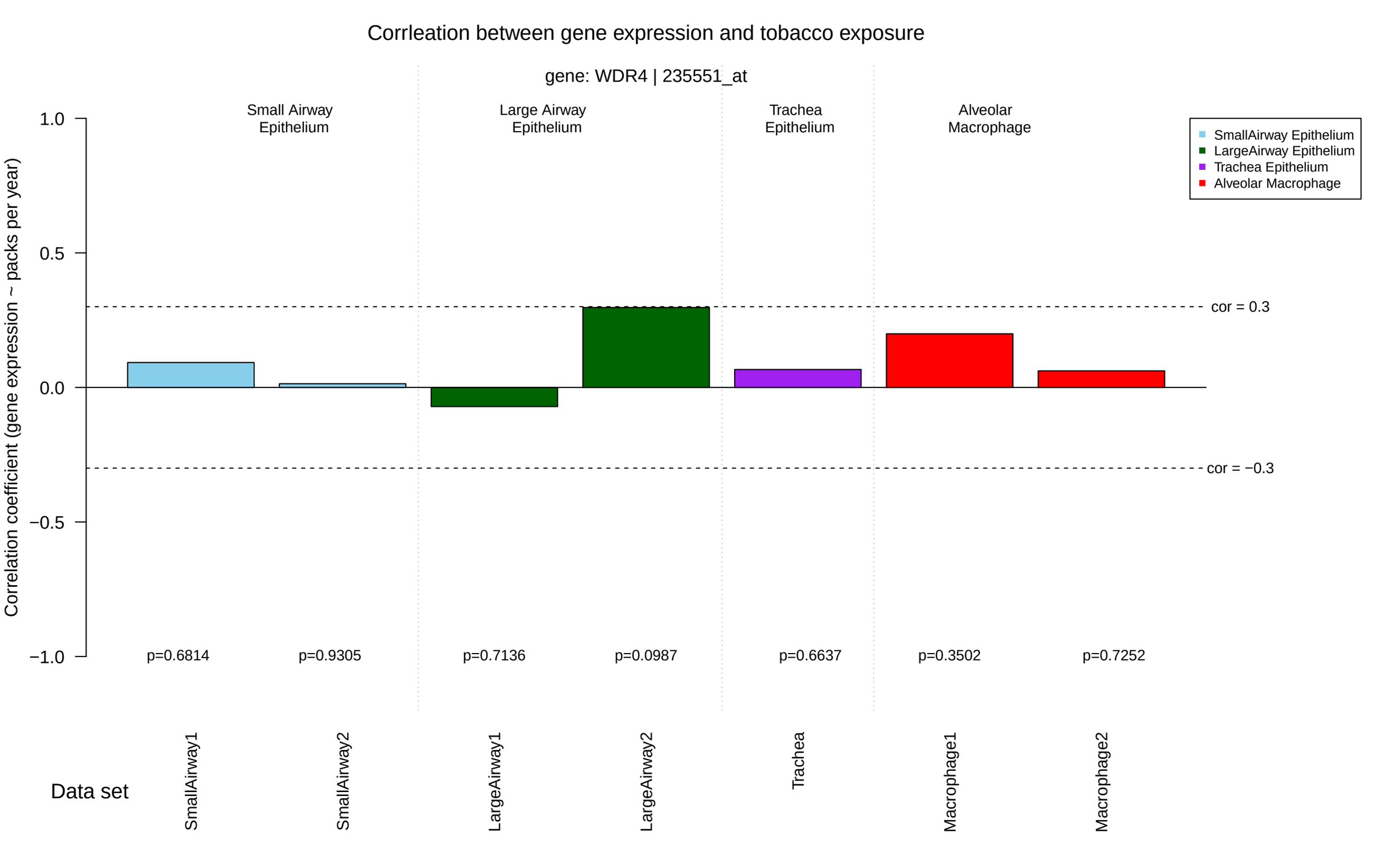

set: LargeAirway1 (p=0.3644)
WDR4 | 235551_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.446)
WDR4 | 235551_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4033)
WDR4 | 235551_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2821)
WDR4 | 235551_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3263)
WDR4 | 235551_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WDR4 | 235551_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6814
p=0.9305
p=0.7136
p=0.0987
p=0.6637
p=0.3502
p=0.7252
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2